



PSO201/PHY204: Assignment 6

Jan-Apr 2025

1. Let φ_0 be the normalized ground state of the harmonic oscillator, such that $\hat{a}\varphi_0 = 0$, where $\hat{a}$ is the annihilation operator and $\hat{a}^\dagger$ is the creation operator.
 - a) Show that the norm of the unnormalized energy eigenstate $\hat{a}^{\dagger n}\varphi_0$ is $\sqrt{n!}$.
 - b) As a consequence, the normalized energy eigenstates take the form, $\varphi_n = \frac{1}{\sqrt{n!}}\hat{a}^{\dagger n}\varphi_0$. Verify that these eigenstates are orthonormal.
 - c) Show that $\hat{a}\varphi_n = \sqrt{n}\varphi_{n-1}$.
 - d) Define the “number” operator $\hat{N} = \hat{a}^\dagger\hat{a}$. What are the eigenfunctions of $\hat{N}$, and what are its eigenvalues?
 - e) Express $\hat{x}$ and $\hat{p}$ in terms of $\hat{a}$ and $\hat{a}^\dagger$. Use these expressions to find $\langle\hat{x}\rangle$ and $\langle\hat{p}\rangle$, if at anytime the harmonic oscillator is in an energy eigenstate φ_n .
 - f) Express $\hat{x}^2$ and $\hat{p}^2$ in terms of $\hat{a}$ and $\hat{a}^\dagger$. Use these expressions to find $\Delta x\Delta p$ when the harmonic oscillator is in an energy eigenstate φ_n .

2. A particle of mass m in a harmonic oscillator potential $V(x) = \frac{1}{2}m\omega^2x^2$ has an initial wave function

$$\psi(x, 0) = \frac{1}{\sqrt{2}}[\varphi_0(x) + i\varphi_1(x)]$$

where φ_0 and φ_1 are the $n = 0$ and $n = 1$ normalized eigenstates.

- a) Write down $\psi(x, t)$ and $|\psi(x, t)|^2$.
 - b) Find the expectation value $\langle\hat{x}\rangle$ as a function of time t . Notice that it oscillates with time. What is the amplitude of the oscillation? What is its angular frequency?
 - c) Find the expectation value $\langle\hat{p}\rangle$ as a function of time.
 - d) Show that the probability distribution of a particle in a harmonic oscillator potential returns to its original shape after the classical period $T = 2\pi/\omega_0$. You should prove this for any harmonic oscillator state, including non-stationary states. What feature of the harmonic oscillator makes this true?
3. Using the creation and annihilation operators find the wavefunctions corresponding to the ground state and the 1st excited state of a harmonic oscillator in position space.
 4. For a particle of mass m in a one-dimensional harmonic oscillator potential, $V(x) = \frac{1}{2}m\omega^2x^2$ write down the most general solution to the time dependent Schrodinger equation, $\psi(x, t)$, in terms of harmonic oscillator eigenstates. Show that the expectation value $\langle\hat{x}\rangle$ as a function of time can be written as $A \cos \omega t + B \sin \omega t$, where A and B are constants. Show explicitly that the time average of the potential energy of a harmonic oscillator in state $\psi(x, t)$, satisfies $\langle V \rangle = \frac{1}{2}\langle E \rangle$.
 5. The Hamiltonian for a harmonic oscillator can be written in dimensionless units ($m = \omega = \hbar = 1$) as $H = \left(\hat{a}^\dagger\hat{a} + \frac{1}{2}\right)$, where $\hat{a} = \frac{1}{\sqrt{2}}(\hat{x} + i\hat{p})$ and $\hat{a}^\dagger = \frac{1}{\sqrt{2}}(\hat{x} - i\hat{p})$. One unnormalized energy eigenfunction is given by $\psi_a = (2x^3 - 3x)e^{-x^2/2}$. Find two other (unnormalized) eigenfunctions which are closest in energy to ψ_a .

6. At time $t = 0$, a particle in the potential $V(x) = \frac{1}{2}m\omega^2 x^2$ is described by the wave function $\psi(x, 0) = A \sum_n \left(\frac{1}{\sqrt{2}}\right)^n \psi_n(x)$ where $\psi_n(x)$ are normalized eigenstates with energy eigenvalues $E_n = (n + \frac{1}{2})\hbar\omega$.
- Find the normalization constant A .
 - Write an expression for $\psi(x, t)$ for $t > 0$.
 - Show that $|\psi(x, t)|^2$ is a periodic function of t and indicate the longest period τ .
 - Find the expectation value of the energy at $t = 0$.
7. Find the energy of the ground state and the first excited quantum state of a particle in the two-dimensional potential $V(x, y) = \frac{1}{2}m\omega^2(x^2 + 4y^2)$.
8. Let $|0\rangle$ and $|1\rangle$ denote the normalized eigenstates corresponding to the ground and first excited states of a one dimensional harmonic oscillator. Find the uncertainty product $\Delta x \Delta p$ in the state $|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$.
9. Consider a simple harmonic oscillator in 1D. Construct a linear combination of $|0\rangle$ and $|1\rangle$ such that $\langle \hat{x} \rangle$ is maximized. Now let the resulting state evolve with time. Find the expectation value $\langle \hat{x} \rangle$ as a function of time.
10. Find the energy eigenvalues of a particle in the potential $V(x) = \frac{1}{2}m\omega^2 x^2 - ax$